# FlightOps DEPI Project — Final DevOps Execution Timeline

**Team Roles & Task Division**

| Status | Primary Deliverables | Responsibility Area | Role |
|---|---|---|---|
| **Completed** | Python Ingestion Script (ingest.py / app.py), OpenSky & Weather APIs integration, SQLite setup. | Data & Application Logic | **Member 1** |
| **Active** | Dockerfile, Image build, Kubernetes Manifests (Deployment, Service, ConfigMap, PVC). | Containerization & Orchestration | **Member 2** |
| Upcoming | Terraform (Cloud Infrastructure), Ansible (Software Provisioning), K8s Deployment on Cloud. | IaC & Cloud Deployment | **Member 3** |
| Upcoming | GitHub Actions / Jenkins Pipelines, Automated Linting, Build, Push, and K8s Deployment Triggers. | Automation & CI/CD | **Member 4** |
| Upcoming | Prometheus, Grafana, Loki Setup, Cluster | Observability & Monitoring | **Member 5** |

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| | Metrics, Application & Log Dashboards. | | |

## Detailed Execution Schedule (Timeline Table)

| Allocated Time | Handover To / Dependency | Key Tasks & Requirements | Responsible Member | Phase / Period |
|---|---|---|---|---|
| ~2 Weeks | **DEPI Deliverable #1 (Oct 16)** | Finalize DEPI Phase 1 Documents: Project Planning, Literature Review, & Requirements Gathering. | **All Members** | **Phase 1**<br><br>*(Now – Oct 16)* |
| 3 Weeks | Handover manifests & image specs to **Member 4** & **Member 3**. | Create optimized Dockerfile, test container locally, write K8s Manifests (Deployment, ConfigMap, PVC). | **Member 2** | **Phase 2.1**<br><br>*(Now – Oct 23)* |
| ~1.5 Weeks | Handover provisioned Cloud Environment to **Member 5** / Application Deployment. | Write Terraform scripts for Cloud VMs/Cluster and Ansible Playbooks for Docker/K8s setup. *(Parallel Execution)* | **Member 3** | **Phase 2.2**<br><br>*(Oct 16 – Oct 25)* |

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| ~2 Weeks | Handover active Pipeline to **Member 3** & **Member 5**. | Build CI/CD Pipeline (GitHub Actions) for automatic Build, Image Push to GHCR/Docker Hub, and Deployment to K8s. | **Member 4** | **Phase 3.1**<br><br>*(Oct 23 – Nov 05)* |
| ~2.5 Weeks | Handover live application environment to **Member 5**. | Deploy the FlightOps service onto the Cloud K8s cluster using the manifests and CI/CD pipelines. | **Member 3** | **Phase 3.2**<br><br>*(Oct 25 – Nov 10)* |
| ~2 Weeks | Handover fully monitored cluster to the **Whole Team**. | Deploy Prometheus, Grafana, and Loki. Build dashboards for CPU/RAM usage, API ingestion metrics, and logs. | **Member 5** | **Phase 4**<br><br>*(Nov 05 – Nov 20)* |
| 4 Days | **System Completed (Nov 24)** | Conduct End-to-End Integration Testing. Fix minor bugs and freeze all code updates (**Internal Code** | **All Members** | **Phase 5**<br><br>*(Nov 20 – Nov 24)* |

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|---|---|---|---|---|
| | | **Freeze**). | | |

## 🎯 Final Preparation Window (Nov 24 – Dec 04, 2026)